

Hospital Management System – SQL Project Report

This project aimed to build and analyze a Hospital Management System using SQL. The database included information about hospitals, doctors, patients, appointments, prescriptions, medications, departments, and rooms. Proper primary and foreign key relationships were created to ensure data accuracy and consistency.

After inserting the sample data, basic queries were used to explore key information. The analysis showed that appointment numbers were highest during mid-year months, and most Neurology rooms had a capacity of two to four beds. Around one-fourth of medications focused on pain or infection treatment, and a few doctors had not yet prescribed any medicines. Many patients had more than one prescription, reflecting repeat or chronic cases.

Advanced SQL queries provided deeper insights. Doctors were ranked by the number of appointments within each hospital, and it was found that adults formed the majority of cardiology consultations and emergency visits. Some hospitals had fewer doctors, indicating staffing gaps, while others like **Green Valley Medical Center** had more cardiology rooms and specialists. The difference between returning patients' visits averaged about 40–50 days, showing regular follow-ups.

Additional insights showed that consultations and checkups made up most of the appointments, with cardiology being one of the busiest specialties. Average room capacity across hospitals was around two to three beds. Overall, the project demonstrated how structured SQL queries can help hospitals monitor operations, identify resource needs, and make better management decisions.